AWS Troubleshooting — Top 50 Interview Questions & Answers

AWS IAM Policies & Permissions — Important Interview Questions & Answers

1.What is an IAM Policy?
An IAM Policy is a JSON document that defines what actions a user, role, or service is allowed or denied to perform on AWS resources.
Simple ga gurthupettuko:
IAM Policy ante **evariki, emi cheyyadaniki permission undho define chese rules**.
Eng: An IAM Policy defines what actions are allowed or denied on AWS resources.

2.What are the main elements of an IAM Policy?
The main elements are Effect, Action, Resource, and Principal in policies where Principal is applicable.

- Effect — Allow or Deny
- Action — What operation is allowed
- Resource — On which resource
- Principal — Who can access the resource

Simple ga gurthupettuko:
Effect → Action → Resource → Principal
Eng: IAM policies define the effect, action, resource, and applicable principal.

3.What is the difference between Allow and Deny in IAM?
Allow gives permission to perform an action, while Deny explicitly blocks the action.
An explicit Deny always takes priority over an Allow.
Simple ga gurthupettuko:
Allow unna, Explicit Deny unte → Deny win.
Eng: An explicit Deny always overrides an Allow.

4.What is an Identity-based Policy?
An Identity-based Policy is attached to an IAM user, group, or role.
It defines what actions that identity can perform on AWS resources.
Simple ga gurthupettuko:
User / Group / Role ki attach chese policy = **Identity-based Policy**.
Eng: An identity-based policy is attached to a user, group, or role to define permissions.

5.What is a Resource-based Policy?
A Resource-based Policy is attached directly to an AWS resource and defines who can access that resource and what actions they can perform.
Examples include S3 bucket policies and SNS topic policies.
Simple ga gurthupettuko:
Resource meeda direct ga attach chese policy = **Resource-based Policy**.
Eng: A resource-based policy is attached to a resource and controls access to that resource.

6.What is the difference between Identity-based and Resource-based Policies?
Identity-based policies are attached to users, groups, or roles.
Resource-based policies are attached directly to resources.
For example, an IAM role can have an identity policy, while an S3 bucket can have a bucket policy.
Simple ga gurthupettuko:
Identity Policy → User/Group/Role
Resource Policy → Resource
Eng: Identity policies are attached to identities, while resource policies are attached to resources.

7.What is an Explicit Deny?
An Explicit Deny is a policy statement that directly denies a specific action or access.
It takes priority over any Allow permission.
For example, even if a user has permission to access an S3 bucket, an explicit Deny can block that access.
Simple ga gurthupettuko:
Allow unna kuda Explicit Deny unte access undadu.
Eng: An explicit Deny overrides an Allow permission.

8.What is the Principle of Least Privilege?
The Principle of Least Privilege means giving only the permissions required to perform a specific task and nothing more.
For example, if an application only needs to read S3 objects, we should give read permission instead of full S3 access.
Simple ga gurthupettuko:
Required permission matrame ivvali, full access ivvakudadhu.
Eng: Least privilege means giving only the permissions required for the task.

9.What is an IAM Role?
An IAM Role is an identity that provides temporary permissions to AWS services, applications, or users.
For example, we can attach an IAM role to an EC2 instance so that the application can access S3 without storing access keys.

Simple ga gurthupettuko:
IAM Role ante **temporary permissions ivvadaniki use chese identity**.
Eng: An IAM Role provides permissions without requiring long-term access keys.

10.Why do we use IAM Roles instead of Access Keys for EC2?
We use IAM Roles because they provide temporary credentials and avoid storing long-term access keys on the EC2 instance.
This is more secure and easier to manage.
Simple ga gurthupettuko:
EC2 → IAM Role better than Access Keys.
Eng: IAM Roles provide temporary credentials and are more secure than storing long-term access keys.

11.An EC2 instance cannot access S3. How will you troubleshoot it?
First, I would check whether the EC2 instance has the correct IAM role attached.
Then I would check the role policy for the required S3 actions, the bucket policy, KMS permissions if encryption is used, and any explicit Deny.
Simple ga gurthupettuko:
EC2 Role → IAM Policy → Bucket Policy → KMS → Explicit Deny
Eng: I would check the EC2 role, permissions, bucket policy, KMS access, and explicit denies.

12.A user has S3 permission but still gets Access Denied. What will you check?
I would check all policies affecting the request, not just the user's IAM policy.
I would check the S3 bucket policy, explicit Deny, permission boundary, SCP if AWS Organizations is used, and KMS permissions if required.
Simple ga gurthupettuko:
IAM → Bucket Policy → Boundary → SCP → KMS → Explicit Deny
Eng: I would check all applicable policies and look for an explicit Deny.

13.What is a Permission Boundary?
A Permission Boundary defines the maximum permissions that an IAM user or role can have.
Even if an identity policy allows more permissions, the identity cannot use permissions outside the boundary.
Simple ga gurthupettuko:
Permission Boundary ante **maximum permission limit**.
Eng: A Permission Boundary defines the maximum permissions an IAM identity can receive.

14.What is an SCP in AWS?
A Service Control Policy (SCP) is used in AWS Organizations to control the maximum permissions available to accounts or organizational units.
An SCP does not directly grant permissions. The required permission must also be allowed by IAM policies.
Simple ga gurthupettuko:
SCP = Account-level permission limit.
Eng: An SCP controls the maximum permissions available within AWS accounts.

15.How would you troubleshoot an IAM Access Denied issue?
I would first identify which user or role is making the request and what resource they are trying to access.
Then I would check identity policies, resource policies, permission boundaries, SCPs, and explicit Denies.
I would also use CloudTrail or IAM policy evaluation tools when required to understand why the request was denied.
Simple ga gurthupettuko:
Identity → Policy → Resource Policy → Boundary → SCP → Explicit Deny → CloudTrail
Eng: I would check all applicable permission layers and identify the exact reason for the Access Denied error.

VPC Security Groups & NACLs — Important Interview Questions & Answers

1.What is a Security Group?
A Security Group is a virtual firewall that controls inbound and outbound traffic for resources such as EC2 instances.
Security Groups are stateful, so if an incoming request is allowed, the response traffic is automatically allowed.
Simple ga gurthupettuko:
Security Group ante **EC2 ki firewall laga work chestundi**.
Eng: A Security Group is a stateful virtual firewall for AWS resources.

2.What is a Network ACL?
A Network ACL is a firewall that controls inbound and outbound traffic at the subnet level.
Unlike Security Groups, Network ACLs are stateless, so inbound and outbound traffic must be allowed separately.
Simple ga gurthupettuko:
NACL ante **subnet level firewall**.
Eng: A Network ACL controls traffic at the subnet level.

3.What is the difference between Security Group and NACL?
A Security Group works at the resource level, while a NACL works at the subnet level.
Security Groups are stateful, while NACLs are stateless.

Simple ga gurthupettuko:
SG → Resource level + Stateful
NACL → Subnet level + Stateless
Eng: Security Groups are stateful resource-level firewalls, while NACLs are stateless subnet-level firewalls.

4.What does stateful mean in Security Groups?
Stateful means when traffic is allowed in one direction, the response traffic is automatically allowed.
For example, if port 80 inbound traffic is allowed, the response from the server does not need a separate outbound rule.
Simple ga gurthupettuko:
Request allow chesthe **response automatic ga allow avthundi**.
Eng: Stateful means response traffic is automatically allowed for an allowed connection.

5.What does stateless mean in NACL?
Stateless means inbound and outbound traffic are evaluated separately.
If inbound traffic is allowed, the response traffic must also be allowed using an outbound rule.
Simple ga gurthupettuko:
Request allow ≠ Response automatically allow.
Eng: In a stateless NACL, inbound and outbound traffic need separate rules.

6.An EC2 instance is not reachable. How will you troubleshoot Security Group and NACL?
First, I would check whether the required port is allowed in the Security Group.
Then I would check the NACL inbound and outbound rules, route table, and whether the application is listening on the required port.
Simple ga gurthupettuko:
Port → SG → NACL → Route → Application
Eng: I would check the port, Security Group, NACL, routing, and application.

7.How do you allow SSH access to an EC2 instance securely?
I would allow TCP port 22 only from the required trusted IP or network instead of allowing it from everywhere.
For example, I can allow port 22 from my organization's public IP range.
Simple ga gurthupettuko:
SSH → Port 22 → Only trusted IP, 0.0.0.0/0 avoid cheyyali.
Eng: I would allow SSH only from trusted IP addresses.

8.How do you allow communication between two EC2 instances?
I would configure the Security Groups to allow the required application port from the source instance's Security Group.
Using the Security Group as the source is better than allowing a broad IP range when possible.
Simple ga gurthupettuko:
EC2-A SG → EC2-B SG → Required Port
Eng: I would allow the required port from the source Security Group.

9.Can a Security Group deny traffic?
No. Security Groups only support Allow rules.
If traffic needs to be explicitly denied, we can use a NACL or another appropriate security control.
Simple ga gurthupettuko:
SG = Allow only. Explicit Deny ledu.
Eng: Security Groups support Allow rules but do not support explicit Deny rules.

10.Can a NACL deny traffic?
Yes. NACLs support both Allow and Deny rules.
Rules are evaluated in order based on the rule number, and the first matching rule is applied.
Simple ga gurthupettuko:
NACL = Allow + Deny + Rule order important.
Eng: NACLs support both Allow and Deny rules, evaluated in rule order.

11.What happens if Security Group allows traffic but NACL blocks it?
The traffic will still be blocked.
Both the Security Group and NACL must allow the required traffic for communication to work.
Simple ga gurthupettuko:
SG Allow + NACL Deny = Traffic Block.
Eng: Traffic is blocked if either the Security Group or NACL prevents it.

12.What happens if NACL allows traffic but Security Group blocks it?
The traffic will be blocked because the Security Group must also allow the traffic.
Both layers need to permit the required communication.
Simple ga gurthupettuko:
NACL Allow + SG Block = Traffic Block.
Eng: Traffic is blocked when the Security Group does not allow it.

13.An application works from one EC2 instance but not from another. What will you check?
I would compare the Security Groups attached to both instances.
Then I would check NACLs, subnet routing, network connectivity, and whether the required port is allowed from the source.

Simple ga gurthupettuko:
Working EC2 vs Non-working EC2 → SG → NACL → Route → Port compare.
Eng: I would compare the network and security configuration of both instances.

14.What is a common mistake when configuring NACLs?
A common mistake is allowing inbound traffic but forgetting the required outbound response traffic.
Because NACLs are stateless, both directions need appropriate rules.
Simple ga gurthupettuko:
NACL lo **Inbound allow chesi Outbound marchipovaddu**.
Eng: Because NACLs are stateless, both inbound and outbound rules must be configured correctly.

15.What is the best way to troubleshoot a network access issue?
I would follow the traffic path step by step.
I would check the route table, Security Group, NACL, required port, and application status to identify exactly where the traffic is being blocked.
Simple ga gurthupettuko:
Route → SG → NACL → Port → Application
Eng: I would follow the network path and check routing, security rules, ports, and the application.
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EC2 Troubleshooting — Important Interview Questions & Answers

1.What will you check if an EC2 instance is not reachable?
First, I would check whether the instance is running and passing its status checks.
Then I would check the network path, Security Group, NACL, route table, and whether the required port is open.
Simple ga gurthupettuko:
EC2 State → Status Check → Route → SG → NACL → Port.
Eng: I would check the instance status, network path, security rules, and required port.

2.EC2 is running but SSH is not working. How will you troubleshoot?
I would check whether port 22 is allowed in the Security Group and whether the instance has the correct public or private connectivity.
Then I would check the route, NACL, key pair, username, and SSH service on the instance.
Simple ga gurthupettuko:
SSH → Port 22 → SG → Route → NACL → Key → SSH Service.
Eng: I would check port 22, networking, the key pair, username, and SSH service.

3.EC2 status check is failing. What will you check?
I would check whether the failure is related to the system or instance status check.
Then I would check system logs and EC2 events. If required, I would restart or stop and start the instance based on the issue.
Simple ga gurthupettuko:
Status Check Fail → System/Instance Check → Events → Logs → Recovery.
Eng: I would identify the failed status check and investigate the underlying instance or system issue.

4.EC2 instance is running but the application is not working. What will you check?
I would check whether the application service is running and listening on the expected port.
Then I would check the Security Group, application configuration, logs, and system resources such as CPU and memory.
Simple ga gurthupettuko:
EC2 Running ≠ Application Running.
App → Port → SG → Resources → Logs.
Eng: I would check the application service, port, security rules, resources, and logs.

5.EC2 CPU utilization is 100%. How will you troubleshoot?
I would check CloudWatch CPU metrics and identify when the CPU increased.
Then I would check running processes, application traffic, logs, and whether the instance needs more capacity or Auto Scaling.
Simple ga gurthupettuko:
CPU 100% → CloudWatch → Process → Traffic → Logs → Scale.
Eng: I would identify the process causing high CPU and check whether scaling is required.

6.EC2 memory usage is very high. What will you check?
I would identify which process is consuming memory and check the application logs.
I would also verify whether the instance has enough memory and whether the application has a memory leak or unusual workload.
Simple ga gurthupettuko:
High Memory → Process → Logs → Application → Instance Size.
Eng: I would identify the memory-consuming process and check the application and instance capacity.

7.EC2 disk space is full. How will you troubleshoot?
I would check which filesystem, directory, or log files are consuming the disk space.
I would clean unnecessary data or logs and increase the EBS volume if more storage is required.

Simple ga gurthupettuko:
Disk Full → Find Usage → Clean Logs/Data → Expand EBS.
Eng: I would identify the disk usage, clean unnecessary data, and increase storage if required.

8.EC2 instance cannot connect to RDS. What will you check?
I would check the RDS status and endpoint first.
Then I would check the RDS Security Group, EC2 Security Group, database port, route tables, subnet connectivity, and database credentials.
Simple ga gurthupettuko:
EC2 → RDS Endpoint → Route → SG → DB Port → Credentials.
Eng: I would check the RDS endpoint, routing, Security Groups, port, and credentials.

9.EC2 instance cannot download packages or access the Internet. What will you check?
First, I would identify whether the EC2 is in a public or private subnet.
For a public subnet, I would check the public IP, route to the Internet Gateway, Security Group, and NACL. For a private subnet, I would check the NAT Gateway and private route.
Simple ga gurthupettuko:
Public → Public IP + IGW.
Private → NAT + Route.
Eng: I would check the correct Internet connectivity path based on the subnet type.

10.EC2 instance suddenly stopped or was terminated. How will you find the reason?
I would check EC2 events and CloudTrail to identify whether the action was performed manually or by another AWS service.
If the instance is part of an Auto Scaling Group, I would also check Auto Scaling activity history.
Simple ga gurthupettuko:
Instance Gone → EC2 Events → CloudTrail → ASG Activity.
Eng: I would check EC2 events, CloudTrail, and Auto Scaling activity to find the reason.

11.EC2 is reachable by private IP but not by public IP. What will you check?
I would check whether the instance has a public or Elastic IP and whether the subnet has a route to the Internet Gateway.
Then I would check the Security Group and NACL rules for the required port.
Simple ga gurthupettuko:
Private works + Public fails → Public IP → IGW Route → SG → NACL.
Eng: I would check the public IP, Internet Gateway route, and security rules.

12.EC2 application works locally but users cannot access it. What will you check?
I would check whether the application is listening on the correct network interface and port.
Then I would check the Security Group, NACL, route table, and if an ALB is used, I would check the target group and health check.
Simple ga gurthupettuko:
Local Works → Binding → Port → SG → NACL → ALB if used.
Eng: I would check the application binding, port, network rules, and load balancer configuration.

13.How would you troubleshoot an EC2 instance that is slow?
I would check CloudWatch metrics such as CPU, network, disk, and other available metrics.
Then I would check running processes, application logs, disk space, database performance, and recent changes.
Simple ga gurthupettuko:
Slow EC2 → CPU → Network → Disk → Process → Logs → DB.
Eng: I would use CloudWatch and system/application logs to identify the resource causing the slowdown.

14.EC2 is healthy but suddenly application traffic increased. What will you do?
I would check CloudWatch metrics and application traffic to confirm the increase.
If the workload is expected to continue, I would scale the instance or use Auto Scaling to handle the increased traffic.
Simple ga gurthupettuko:
Traffic ↑ → Monitor → Capacity Check → Scale.
Eng: I would monitor the workload and scale the application if required.

15.What is your overall approach to troubleshooting an EC2 issue?
First, I identify whether the problem is with the instance, network, application, or dependency.
Then I check status checks, CloudWatch metrics, Security Groups, routing, application logs, and recent changes. Finally, I fix the root cause and verify the application.
Simple ga gurthupettuko:
EC2 → Network → Security → Application → Logs → Root Cause → Fix → Verify.
Eng: I identify the failed layer, investigate using metrics and logs, fix the root cause, and verify the result.

ALB / Target Groups / Health Checks — Important Interview Questions & Answers

1.ALb is showing targets as Unhealthy. How will you troubleshoot?
First, I would check the target health reason in the Target Group.
Then I would verify the health check path, port, Security Group, NACL, and whether the application is running on the target instance.
Simple ga gurthupettuko:
Unhealthy → Health Check → Port → SG → NACL → Application.
Eng: I would check the health check configuration, network access, port, and application.

2.ALb returns 502 Bad Gateway. What will you check?
I would check whether the ALB can successfully communicate with the target.
Then I would check the Target Group, target port, application status, and application logs for backend errors.
Simple ga gurthupettuko:
502 → ALB → Target Group → Target Port → Application → Logs.
Eng: I would check the target group, backend port, application, and logs.

3.ALb returns 504 Gateway Timeout. What does it usually mean?
A 504 usually means the ALB did not receive a response from the backend within the expected time.
I would check application response time, database queries, backend performance, and timeout-related settings.
Simple ga gurthupettuko:
504 → Backend response late → App → DB → Timeout.
Eng: I would check why the backend is taking too long to respond.

4.Health check is failing even though the application works from the EC2 instance. Why?
The ALB health check may be using the wrong path or port.
I would check the health check configuration and verify that the application is accessible from the ALB on the configured port.
Simple ga gurthupettuko:
App works → ALB Health Check fail → Path / Port / SG check.
Eng: I would verify the health check path, port, and network access from the ALB.

5.What is a Target Group and how would you troubleshoot it?
A Target Group contains the backend targets, such as EC2 instances, that receive traffic from the ALB.
I would check registered targets, target health, target port, health check configuration, and whether the targets are receiving traffic.
Simple ga gurthupettuko:
Target Group = **Backend servers list + Health Check**.
Eng: A Target Group contains backend targets and performs health checks on them.

6.ALb is working but traffic is not reaching the backend. What will you check?
I would check the ALB listener and listener rules to make sure they forward traffic to the correct Target Group.
Then I would check the target health, target port, and Security Groups.
Simple ga gurthupettuko:
Client → ALB Listener → Rule → Target Group → EC2.
Eng: I would verify the listener, forwarding rule, target group, target health, and security rules.

7.How do you troubleshoot an ALB listener issue?
I would check whether the listener is configured for the correct protocol and port, such as HTTP 80 or HTTPS 443.
Then I would check the listener rules and confirm that traffic is forwarded to the correct Target Group.
Simple ga gurthupettuko:
Listener → Protocol → Port → Rule → Target Group.
Eng: I would check the listener protocol, port, rules, and target group.

8.Only one EC2 target is receiving traffic while other targets are not. What will you check?
I would check the health status of all targets in the Target Group.
If some targets are unhealthy, I would investigate their health check, application port, Security Group, and application status.
Simple ga gurthupettuko:
One Target traffic → All Target Health compare → Unhealthy targets troubleshoot.
Eng: I would compare target health and investigate why some targets are unhealthy.

9.ALb is returning 404. How will you troubleshoot?
I would check the listener rules and the requested URL path.
If path-based routing is configured, I would verify that the request matches the correct rule and is forwarded to the expected Target Group.
Simple ga gurthupettuko:
404 → URL Path → Listener Rule → Target Group → Application.
Eng: I would check the requested path and listener routing rules.

10.HTTPS is not working on the ALB. What will you check?
I would check whether the HTTPS listener is configured on port 443 and whether a valid SSL/TLS certificate is attached.
I would also check the Security Group and listener configuration.

Simple ga gurthupettuko:
HTTPS → Port 443 → Certificate → Listener → SG.
Eng: I would check the HTTPS listener, certificate, port 443, and Security Group.

11.ALb works on HTTP but not HTTPS. What will you check?
I would check the HTTPS listener on port 443 and verify the attached certificate.
Then I would check the certificate validity, Security Group rules, listener configuration, and DNS if applicable.
Simple ga gurthupettuko:
HTTP works + HTTPS fails → **443 → Certificate → Listener → SG → DNS**.
Eng: I would check the HTTPS listener, certificate, Security Group, and DNS configuration.

12.Users are getting intermittent errors from the ALB. How will you troubleshoot?
I would check whether only specific targets are causing the problem.
I would compare target health, application logs, ALB logs, response times, and error rates for each target.
Simple ga gurthupettuko:
Intermittent → Target-wise Health → Logs → Response Time → Errors.
Eng: I would compare individual targets to identify whether one backend is causing the issue.

13.What would you check if ALB health checks are passing but the application still has issues?
A successful health check only confirms that the configured health check endpoint is responding.
I would check the actual application APIs, application logs, database connectivity, CPU, memory, and other dependencies.
Simple ga gurthupettuko:
Health Check Pass ≠ Full Application Healthy.
Eng: A passing health check does not guarantee that the complete application is healthy.

14.How would you troubleshoot an ALB after a new deployment?
I would check whether the new targets are healthy and whether the listener is forwarding traffic correctly.
Then I would check application logs, error rates, response times, and compare the new version with the previous working version.
Simple ga gurthupettuko:
Deployment → Target Health → Listener → Logs → Errors → Compare Version.
Eng: I would check target health, routing, logs, and application behavior after deployment.

15.What is the best approach to troubleshoot an ALB issue?
I would follow the request flow from the client to the ALB and then to the Target Group and backend.
I would check the listener, listener rules, target health, health checks, ports, Security Groups, and backend application.
Simple ga gurthupettuko:
Client → Listener → Rule → Target Group → Health Check → EC2 → Application.
Eng: I would follow the traffic path from the client through the ALB to the backend and identify where it fails.

AWS Auto Scaling — Important Interview Questions & Answers

1.What is Auto Scaling?
Auto Scaling automatically increases or decreases the number of EC2 instances based on application demand and configured rules.
It helps maintain application availability and handle changing traffic.
Simple ga gurthupettuko:
Traffic ekkuva → **Instances increase**
Traffic takkuva → **Instances decrease**
Eng: Auto Scaling automatically adjusts EC2 capacity based on demand.

2.What is an Auto Scaling Group (ASG)?
An Auto Scaling Group is a logical group of EC2 instances that Auto Scaling manages.
It maintains the required number of instances and can replace unhealthy instances automatically.
Simple ga gurthupettuko:
ASG ante **EC2 instances ni manage chese group**.
Eng: An ASG manages a group of EC2 instances and maintains the desired capacity.

3.What are Desired, Minimum, and Maximum capacity?
Desired Capacity — Normal ga maintain cheyyali anukune instance count.
Minimum Capacity — Minimum ga undalsina instances.
Maximum Capacity — Maximum ga create cheyyagalige instances.
For example: Min = 2, Desired = 3, Max = 5.
Simple ga gurthupettuko:
Min = Minimum
Desired = Normal
Max = Maximum
Eng: Minimum, desired, and maximum define the scaling limits and normal capacity.

4.What is a Launch Template?

A Launch Template defines how new EC2 instances should be launched.
It can contain AMI, instance type, Security Group, key pair, and other instance configuration.
Simple ga gurthupettuko:
Launch Template ante **new EC2 ela create avvali ani define chese configuration**.
Eng: A Launch Template defines the configuration used to launch EC2 instances.

5.How does Auto Scaling know when to launch or terminate instances?
Auto Scaling uses scaling policies and CloudWatch metrics to decide when to increase or decrease capacity.
For example, if average CPU usage stays high, the ASG can launch additional instances.
Simple ga gurthupettuko:
CloudWatch Metric → Scaling Policy → ASG → EC2
Eng: Auto Scaling uses metrics and scaling policies to adjust capacity.

6.What is a Target Tracking Scaling Policy?
Target Tracking automatically adjusts the number of instances to maintain a target metric value.
For example, we can configure the ASG to maintain average CPU utilization around 50%.
Simple ga gurthupettuko:
Target Tracking ante **oka target value maintain cheyyadaniki instances automatic ga scale avvadam**.
Eng: Target Tracking adjusts capacity to maintain a target metric value.

7.What is the difference between Scale Out and Scale In?
Scale Out means increasing the number of EC2 instances when demand increases.
Scale In means reducing the number of EC2 instances when demand decreases.
Simple ga gurthupettuko:
Scale Out = EC2 +
Scale In = EC2 –
Eng: Scale Out adds instances, while Scale In removes instances.

8.What happens when an EC2 instance in an ASG becomes unhealthy?
The Auto Scaling Group can terminate the unhealthy instance and launch a replacement instance to maintain the desired capacity.
If an ALB is attached, the ASG can also use load balancer health checks.
Simple ga gurthupettuko:
Unhealthy EC2 → **Terminate → New EC2 launch → Desired count maintain**
Eng: Auto Scaling replaces unhealthy instances to maintain capacity.

9.How does Auto Scaling work with an ALB?
The ASG launches EC2 instances and registers them with the ALB Target Group.
The ALB distributes traffic only to healthy targets.
Simple ga gurthupettuko:
ASG → EC2 → Target Group → ALB → Users
Eng: Auto Scaling manages the instances while the ALB distributes traffic among healthy instances.

10.Auto Scaling is not launching new instances. How will you troubleshoot?
I would check the ASG desired, minimum, and maximum capacity.
Then I would check the scaling policy, CloudWatch metric, Launch Template, subnet availability, IAM permissions, and ASG activity history.
Simple ga gurthupettuko:
Capacity → Policy → CloudWatch → Launch Template → Subnet → Activity
Eng: I would check capacity settings, scaling policies, metrics, launch configuration, and ASG activity.

11.Auto Scaling is launching too many instances. What will you check?
I would check the CloudWatch metric and scaling policy that is triggering the scale-out action.
Then I would check whether the metric threshold is too aggressive or whether the application is receiving unusually high traffic.
Simple ga gurthupettuko:
Too many EC2 → **Metric → Threshold → Policy → Traffic**
Eng: I would check the scaling metric, threshold, policy, and actual workload.

12.EC2 instances are healthy but Auto Scaling keeps replacing them. Why?
I would check the health check type and the health status reported by the ALB if load balancer health checks are enabled.
Then I would check the application health check path, port, Security Group, and application logs.
Simple ga gurthupettuko:
EC2 Healthy → ASG replaces → **Health Check → ALB → Port → App**
Eng: I would check whether the ASG or ALB is incorrectly marking the instances as unhealthy.

13.What is a cooldown period in Auto Scaling?
A cooldown period gives Auto Scaling time to stabilize after a scaling activity before performing another scaling action.
It helps prevent unnecessary repeated scaling.
Simple ga gurthupettuko:
Scale ayyaka **immediately malli scale avvakunda konchem stabilize time**.
Eng: A cooldown period helps prevent repeated scaling actions while the system stabilizes.

14.How would you design Auto Scaling for a highly available application?
I would deploy the Auto Scaling Group across multiple Availability Zones.
I would place an ALB in front of the instances and configure health checks and appropriate scaling policies.
Simple ga gurthupettuko:
Multi-AZ ASG + ALB + Health Check + Scaling Policy
Eng: I would use a Multi-AZ ASG with an ALB, health checks, and scaling policies.

15.How would you troubleshoot an Auto Scaling issue in production?
I would check ASG activity history first to understand what action Auto Scaling is taking.
Then I would check desired/min/max capacity, scaling policies, CloudWatch metrics, instance health, ALB health checks, Launch Template, and recent changes.
Simple ga gurthupettuko:
ASG Activity → Capacity → Policy → CloudWatch → ALB Health → Launch Template → Logs
Eng: I would use ASG activity, CloudWatch, health checks, and configuration details to identify the root cause.

S3 Permissions & Policies — Important Interview Questions & Answers

1.What is an S3 Bucket Policy?
An S3 Bucket Policy is a resource-based policy attached to an S3 bucket.
It defines who can access the bucket and what actions they can perform, such as GetObject or PutObject.
Simple ga gurthupettuko:
Bucket Policy ante **S3 bucket ki access evariki undho define chese policy**.
Eng: A bucket policy controls access to an S3 bucket and its objects.

2.What is the difference between IAM Policy and S3 Bucket Policy?
An IAM Policy is attached to an IAM user, group, or role.
An S3 Bucket Policy is attached directly to the S3 bucket.
Simple ga gurthupettuko:
IAM Policy → User/Group/Role
Bucket Policy → S3 Bucket
Eng: IAM policies are identity-based, while bucket policies are resource-based.

3.A user has S3 permission but gets Access Denied. What will you check?
I would check the user's IAM policy and the S3 bucket policy.
Then I would check for explicit Deny, Permission Boundary, SCP, object ownership, and KMS permissions if the object is encrypted.
Simple ga gurthupettuko:
IAM → Bucket Policy → Explicit Deny → Boundary → SCP → KMS
Eng: I would check all applicable policies and look for an explicit Deny.

4.How do you give an EC2 instance access to an S3 bucket?
I would create an IAM Role with the required S3 permissions and attach the role to the EC2 instance.
I would avoid storing access keys directly on the EC2 instance.
Simple ga gurthupettuko:
EC2 → IAM Role → S3 Permission → Bucket
Eng: I would use an IAM Role with the required S3 permissions.

5.How do you allow access to only one S3 bucket?
I would create a policy that allows only the required S3 actions on the specific bucket and its objects.
I would avoid using broad permissions such as s3:* on all resources.
Simple ga gurthupettuko:
Required Bucket + Required Actions only.
Eng: I would restrict the policy to the required bucket and actions.

6.How do you give read-only access to an S3 bucket?
I would allow only the required read actions, such as s3:GetObject, and avoid write or delete permissions.
The policy should be limited to the required bucket and objects.
Simple ga gurthupettuko:
Read-only → **GetObject**, write/delete permissions vaddu.
Eng: I would grant only the required read permissions.

7.How do you allow one AWS account to access another account's S3 bucket?
I would configure cross-account access using the S3 bucket policy and IAM permissions in the other account.
The requesting identity must have the required permissions, and the bucket policy must allow that access.
Simple ga gurthupettuko:
Account A User/Role → Account B Bucket Policy + IAM Permission
Eng: Cross-account S3 access requires appropriate permissions on both sides.

8.How would you make an S3 bucket private?
I would keep Block Public Access enabled and avoid public bucket policies or ACLs.
I would give access only to required IAM roles, users, or AWS services.
Simple ga gurthupettuko:
Private S3 → Block Public Access + Least Privilege
Eng: I would block public access and allow only required identities.

9.What is S3 Block Public Access?
S3 Block Public Access is a set of controls that helps prevent public access to S3 buckets and objects.
It can block public bucket policies, ACLs, and other public access configurations.
Simple ga gurthupettuko:
Block Public Access ante **accidentally S3 public avvakunda protect chese safety layer**.
Eng: Block Public Access helps prevent unintended public access to S3.

10.What is S3 Object Ownership?
S3 Object Ownership controls ownership of objects uploaded to a bucket.
With **Bucket owner enforced**, ACLs are disabled and the bucket owner owns all objects.
Simple ga gurthupettuko:
Bucket owner enforced → **Bucket owner owns objects + ACLs disabled.**
Eng: Object Ownership determines who owns uploaded objects and can disable ACLs.

11.An application can list an S3 bucket but cannot download objects. Why?
Listing a bucket and reading objects are different permissions.
The application may have s3:ListBucket permission but not s3:GetObject.
Simple ga gurthupettuko:
ListBucket = Bucket list
GetObject = Object read
Eng: The application needs s3:GetObject to read objects.

12.An application can upload to S3 but cannot download. What will you check?
I would check whether the role or user has s3:PutObject but is missing s3:GetObject.
I would also check bucket policies, explicit Deny, and KMS permissions if encryption is used.
Simple ga gurthupettuko:
Upload works → PutObject exists.
Download fail → **GetObject / KMS / Deny check.**
Eng: I would check GetObject permission, bucket policies, and KMS access.

13.An S3 object is encrypted with KMS and the application gets Access Denied. What will you check?
I would check whether the application's IAM role has the required KMS permissions, such as kms:Decrypt.
I would also check the KMS key policy and whether the correct KMS key is being used.
Simple ga gurthupettuko:
S3 encrypted → **S3 Permission + KMS Permission + Key Policy**.
Eng: I would check both S3 permissions and KMS permissions.

14.How would you follow the Principle of Least Privilege for S3?
I would give only the required actions on only the required resources.
For example, if an application only needs to read files from one bucket, I would allow s3:GetObject only for that bucket's required object path.
Simple ga gurthupettuko:
Required Action + Required Bucket + Required Path only.
Eng: I would restrict S3 permissions to the minimum required actions and resources.

15.How would you troubleshoot an S3 Access Denied issue in production?
First, I would identify which IAM user or role is making the request.
Then I would check IAM policy, bucket policy, Block Public Access, Object Ownership, Permission Boundary, SCP, KMS permissions, and CloudTrail to understand why the request was denied.
Simple ga gurthupettuko:
Identity → IAM → Bucket Policy → Public Access → Boundary → SCP → KMS → CloudTrail
Eng: I would check all permission layers and use CloudTrail when necessary to identify the root cause.

AWS RDS Connectivity & Performance — Important Interview Questions & Answers

1.EC2 cannot connect to RDS. How will you troubleshoot?
First, I would check whether the RDS instance is available and verify the correct RDS endpoint and port.
Then I would check the Security Groups, NACLs, route tables, subnet configuration, and database credentials.
Simple ga gurthupettuko:
RDS Status → Endpoint → Port → SG → NACL → Route → Credentials
Eng: I would check the RDS status, endpoint, port, networking, security rules, and credentials.

2.What is an RDS endpoint?
An RDS endpoint is the DNS address used by applications to connect to the RDS database.
Applications use the endpoint instead of directly using the database server's IP address.
Simple ga gurthupettuko:
RDS Endpoint ante **application database ki connect ayye address**.
Eng: An RDS endpoint is the DNS address used to connect to an RDS database.

3.RDS is running but the application cannot connect. What will you check?
I would verify that the application is using the correct endpoint, port, database name, username, and password.
Then I would check Security Groups, subnet connectivity, and whether the database is accepting connections.
Simple ga gurthupettuko:
Endpoint → Port → Credentials → SG → Network → DB
Eng: I would verify the connection details, networking, security rules, and database availability.

4.What is the default port for MySQL and PostgreSQL in RDS?
MySQL commonly uses port **3306**.
PostgreSQL commonly uses port **5432**.
Simple ga gurthupettuko:
MySQL → 3306
PostgreSQL → 5432
Eng: MySQL uses port 3306 and PostgreSQL uses port 5432 by default.

5.How do you securely allow EC2 to connect to RDS?
I would keep RDS private and allow the database port in the RDS Security Group only from the EC2 application's Security Group.
I would avoid allowing the database port from 0.0.0.0/0.
Simple ga gurthupettuko:
EC2 SG → RDS SG → DB Port
Eng: I would allow database access only from the application's Security Group.

6.RDS CPU utilization is very high. How will you troubleshoot?
I would check CloudWatch metrics and identify when the CPU increased.
Then I would check database connections, slow queries, workload, application traffic, and Performance Insights if available.
Simple ga gurthupettuko:
High CPU → CloudWatch → Queries → Connections → Traffic
Eng: I would investigate database workload, queries, connections, and resource usage.

7.RDS storage is almost full. What will you do?
I would check the storage usage and identify what is consuming the space.
Then I would clean unnecessary data if appropriate and increase the allocated storage or configure storage autoscaling if supported and suitable.
Simple ga gurthupettuko:
Storage Full → **Find Usage → Clean → Increase/Autoscale**
Eng: I would investigate storage usage and increase or automatically scale storage when appropriate.

8.RDS database is responding slowly. How will you troubleshoot?
I would check CPU, memory-related metrics, storage performance, database connections, and query performance.
I would also check slow queries and use Performance Insights to identify database load if enabled.
Simple ga gurthupettuko:
Slow DB → CPU → Storage → Connections → Slow Queries → Performance Insights
Eng: I would check resource usage and identify slow or expensive database queries.

9.What is RDS Multi-AZ?
Multi-AZ creates a standby database instance in another Availability Zone for high availability.
If the primary database fails, RDS can automatically fail over to the standby.
Simple ga gurthupettuko:
Primary AZ → Standby AZ → Failure → Automatic Failover
Eng: Multi-AZ provides high availability through a standby instance in another Availability Zone.

10.What is the difference between Multi-AZ and Read Replica?
Multi-AZ is mainly used for high availability and failover.
Read Replicas are mainly used to scale read workloads by providing additional read-only database copies.
Simple ga gurthupettuko:
Multi-AZ → HA / Failover
Read Replica → Read Scaling
Eng: Multi-AZ is for high availability, while Read Replicas are mainly for read scaling.

11.RDS connection count is very high. What will you check?
I would check whether application traffic has increased and whether connections are being properly managed.
I would check connection pooling, long-running connections, application behavior, and the database's connection limits.
Simple ga gurthupettuko:
High Connections → **Traffic → Connection Pool → Long Connections → DB Limit**
Eng: I would check traffic, connection pooling, long-running connections, and database limits.

12.Application gets connection timeout while connecting to RDS. What will you check?
I would check the RDS endpoint and port first.
Then I would check Security Groups, NACLs, route tables, subnet connectivity, and whether the database is reachable from the application subnet.
Simple ga gurthupettuko:
Timeout → **Endpoint → Port → SG → NACL → Route → Subnet**
Eng: I would check the network path and security configuration between the application and RDS.

13.How would you troubleshoot a sudden RDS performance issue?
I would compare current CloudWatch metrics with the previous normal period.
Then I would check CPU, connections, storage, database load, slow queries, application traffic, and recent configuration or application changes.
Simple ga gurthupettuko:
Sudden Slow → **Compare Metrics → DB Load → Queries → Traffic → Recent Changes**
Eng: I would compare metrics and investigate database load, queries, traffic, and recent changes.

14.What happens when an RDS Multi-AZ primary database fails?
RDS automatically performs a failover to the standby instance in another Availability Zone.
The RDS endpoint remains the same, so the application normally does not need to change its database endpoint.
Simple ga gurthupettuko:
Primary Fail → **Standby becomes Primary → Same Endpoint**
Eng: RDS automatically fails over to the standby while keeping the database endpoint unchanged.

15.What is your overall approach to troubleshoot an RDS issue?
First, I identify whether the issue is connectivity, performance, storage, or availability related.
Then I check RDS status, endpoint, port, Security Groups, networking, CloudWatch metrics, database connections, queries, logs, and recent changes.
Simple ga gurthupettuko:
Connectivity → Security → Network → Resources → Queries → Logs → Root Cause
Eng: I identify the problem area first and then use networking, metrics, logs, and database information to find the root cause.

AWS CloudWatch / Logs / Alarms — Important Interview Questions & Answers

1.What is Amazon CloudWatch?
Amazon CloudWatch is a monitoring and observability service used to monitor AWS resources, applications, logs, and performance metrics.
It helps us identify issues and take actions based on monitoring data.
Simple ga gurthupettuko:
CloudWatch ante **AWS resources and applications ni monitor cheyyadaniki use chese service**.
Eng: CloudWatch is used to monitor AWS resources, applications, metrics, and logs.

2.What is a CloudWatch Metric?
A Metric is a time-ordered set of data points used to measure the performance or behavior of a resource or application.
Examples are EC2 CPU utilization, network traffic, and RDS database connections.
Simple ga gurthupettuko:
Metric ante **system performance ni measure chese value**.
Eng: A CloudWatch Metric is a value used to measure resource or application performance.

3.What is a CloudWatch Alarm?
A CloudWatch Alarm monitors a metric and performs an action when the metric crosses a configured threshold.
For example, an alarm can trigger when EC2 CPU utilization stays above 80%.
Simple ga gurthupettuko:
Metric → Threshold cross → Alarm → Action
Eng: A CloudWatch Alarm monitors a metric and triggers an action when a condition is met.

4.How does CloudWatch Alarm work?
First, we configure a metric, threshold, evaluation period, and action.
When the metric meets the configured condition for the required evaluation period, the alarm changes state and can trigger an action such as an SNS notification.
Simple ga gurthupettuko:
Metric → Threshold → Evaluation → Alarm → SNS/Action
Eng: An alarm evaluates a metric against a threshold and can trigger an action.

5.What is the difference between CloudWatch Metrics and Logs?
Metrics are numerical values used to monitor performance.
Logs contain detailed application or system information such as errors, events, and messages.
Simple ga gurthupettuko:
Metrics = Numbers
Logs = Detailed information
Eng: Metrics provide numerical monitoring data, while logs provide detailed event and application information.

6.What is a CloudWatch Log Group?
A Log Group is a collection of related log streams that share common retention and monitoring settings.
For example, application logs from multiple instances can be stored in a common log group.
Simple ga gurthupettuko:
Log Group ante **related logs ni organize chese container**.
Eng: A Log Group is a container for related log streams.

7.What is a CloudWatch Log Stream?
A Log Stream is a sequence of log events from a specific source.
For example, logs from a particular EC2 instance or application process can be stored in a log stream.
Simple ga gurthupettuko:
Log Group = Container
Log Stream = Individual log source
Eng: A Log Stream contains log events from a specific source.

8.EC2 CPU is high. How will you troubleshoot using CloudWatch?
I would check the CPU utilization metric and identify when the increase started.
Then I would check other metrics, application logs, and running processes to identify what is causing the high CPU.
Simple ga gurthupettuko:
CPU Metric → Time → Logs → Process → Root Cause
Eng: I would use CloudWatch metrics and logs to identify the cause of high CPU usage.

9.Application is down but EC2 is healthy. How will you troubleshoot?
I would check application logs in CloudWatch Logs and verify whether the application process is running.
Then I would check the application port, ALB health if applicable, CPU/memory, recent deployments, and errors.
Simple ga gurthupettuko:
EC2 Healthy ≠ App Healthy
Logs → Process → Port → ALB → Recent Changes.
Eng: I would check application logs, process status, ports, load balancer health, and recent changes.

10.How can CloudWatch automatically notify when CPU is high?
I would create a CloudWatch Alarm on the EC2 CPU utilization metric.
For example, if CPU remains above 80% for a configured period, the alarm can send a notification through SNS.
Simple ga gurthupettuko:
CPU > 80% → CloudWatch Alarm → SNS → Notification
Eng: A CloudWatch Alarm can monitor CPU and send an SNS notification when the threshold is reached.

11.What is CloudWatch Logs Insights?
CloudWatch Logs Insights is used to interactively search and analyze log data using queries.
It is useful for finding errors, specific request IDs, status codes, or patterns in large amounts of logs.
Simple ga gurthupettuko:
Logs ekkuva unte **query petti required error/data fast ga find cheyyadaniki** Logs Insights use chestham.
Eng: Logs Insights helps query and analyze large amounts of log data.

12.How would you troubleshoot an application error using CloudWatch Logs?
First, I would identify the relevant Log Group and Log Stream.
Then I would search for error messages, timestamps, request IDs, and stack traces using Logs Insights if required.
Simple ga gurthupettuko:
Log Group → Log Stream → Timestamp → Error → Request ID → Root Cause
Eng: I would search the relevant logs for errors and trace the request to identify the root cause.

13.CloudWatch Alarm is not triggering. What will you check?
I would check whether the alarm is monitoring the correct metric, threshold, period, and evaluation settings.
Then I would check whether the metric is receiving data and whether the alarm action is configured correctly.

Simple ga gurthupettuko:
Alarm Not Triggering → **Metric → Data → Threshold → Period → Action**
Eng: I would verify the metric data, threshold, evaluation settings, and alarm action.

14.CloudWatch shows no data for a metric. What could be the reason?
The resource may not be sending the expected metric, or the metric configuration may be incorrect.
I would check the metric namespace, dimensions, monitoring configuration, agent if applicable, and the selected time range.
Simple ga gurthupettuko:
No Data → **Metric → Namespace → Dimensions → Agent → Time Range**
Eng: I would verify the metric configuration and whether the required monitoring data is being published.

15.How would you use CloudWatch for production troubleshooting?
I would first check important metrics such as CPU, memory where available, network, latency, errors, and database metrics.
Then I would correlate those metrics with application logs and recent changes to identify the root cause.
Simple ga gurthupettuko:
Metrics → Logs → Errors → Recent Changes → Root Cause
Eng: I would correlate metrics and logs with recent changes to identify and resolve production issues.

AWS Lambda Troubleshooting — Important Interview Questions & Answers

1.Lambda function is failing. How will you troubleshoot?
First, I would check the Lambda invocation errors and CloudWatch Logs.
Then I would check the function code, input event, environment variables, IAM permissions, timeout, and dependencies.
Simple ga gurthupettuko:
Lambda Error → CloudWatch Logs → Event → IAM → Config → Timeout
Eng: I would check logs, input, permissions, configuration, and timeout settings.

2.Lambda function is timing out. What will you check?
I would check the configured Lambda timeout and CloudWatch execution duration.
Then I would check slow database/API calls, network connectivity, external services, and whether the function is doing unnecessary processing.
Simple ga gurthupettuko:
Timeout → Duration → DB/API → Network → Code
Eng: I would identify what is taking too long and check external dependencies and code execution.

3.Lambda is getting Access Denied. How will you troubleshoot?
I would identify which AWS action the Lambda function is trying to perform.
Then I would check the Lambda execution role, IAM policy, resource policy if applicable, KMS permissions, and any explicit Deny.
Simple ga gurthupettuko:
Lambda → Execution Role → IAM Policy → Resource Policy → KMS → Deny
Eng: I would check the execution role and all applicable permission policies.

4.Lambda cannot access RDS. What will you check?
If Lambda is configured inside a VPC, I would check the subnet configuration, route tables, Security Groups, and RDS Security Group.
I would also verify the RDS endpoint, port, credentials, and whether the Lambda has the required network access.
Simple ga gurthupettuko:
Lambda VPC → Route → SG → RDS SG → Port → Endpoint
Eng: I would check VPC networking, Security Groups, the RDS endpoint, and database access.

5.Lambda cannot access the Internet after configuring it inside a VPC. Why?
A Lambda function attached to a VPC does not automatically get Internet access.
For outbound Internet access, the Lambda subnets need an appropriate route through a NAT Gateway.
Simple ga gurthupettuko:
VPC Lambda → Private Subnet → NAT Gateway → Internet
Eng: VPC-connected Lambda requires appropriate VPC networking, such as a NAT Gateway, for outbound Internet access.

6.Lambda is being invoked but the expected result is not produced. What will you check?
I would check the incoming event and verify whether the Lambda is receiving the expected input.
Then I would check the function logs, code logic, environment variables, IAM permissions, and downstream services.
Simple ga gurthupettuko:
Event → Logs → Code → Environment → IAM → Dependency
Eng: I would verify the event, code execution, configuration, permissions, and dependencies.

7.Lambda has high error rates. How will you troubleshoot?
I would check CloudWatch metrics such as Errors, Invocations, Duration, and Throttles.
Then I would check CloudWatch Logs to identify the exact error and correlate it with recent deployments or configuration changes.
Simple ga gurthupettuko: **Errors → Metrics → Logs → Recent Deployment → Root Cause**
Eng: I would use Lambda metrics and logs to identify the source of the errors.

8.What is Lambda throttling?
Lambda throttling happens when the function reaches its available concurrency limit and new invocations cannot be processed normally.
I would check the function's concurrency configuration and invocation rate.
Simple ga gurthupettuko:
Requests ekkuva → Concurrency limit reach → Throttling
Eng: Throttling occurs when Lambda cannot accept additional invocations because of concurrency limits.

9.Lambda is getting throttled. How will you troubleshoot?
I would check the invocation rate and current concurrency.
Then I would check reserved concurrency, account concurrency limits, traffic spikes, and whether the function needs better concurrency management.
Simple ga gurthupettuko:
Throttle → Invocations → Concurrency → Reserved Concurrency → Traffic
Eng: I would check concurrency limits, invocation rate, and traffic patterns.

10.Lambda function is working in the console but failing when invoked by API Gateway. Why?
The event received from API Gateway may be different from the test event used in the console.
I would check the API Gateway integration, Lambda permissions, event structure, and CloudWatch logs.
Simple ga gurthupettuko:
Console Works → API Gateway Event/Integration → Permission → Logs
Eng: I would compare the API Gateway event with the working test event and check integration permissions.

11.Lambda is taking longer after a new deployment. How will you troubleshoot?
I would compare the new version's duration and error metrics with the previous version.
Then I would check the code changes, external API calls, database queries, configuration, and CloudWatch Logs.
Simple ga gurthupettuko:
New Deployment → Duration Compare → Code → DB/API → Logs
Eng: I would compare the new and previous versions and investigate the change causing higher execution time.

12.What is a Lambda Cold Start?
A cold start happens when Lambda needs to create a new execution environment before running the function.
This can add some startup latency, especially for functions with large dependencies or certain runtime configurations.
Simple ga gurthupettuko:
New execution environment create avvali → **Extra startup time = Cold Start**.
Eng: A cold start is the startup time required to initialize a new Lambda execution environment.

13.Lambda works sometimes but fails sometimes. How will you troubleshoot?
I would check CloudWatch logs and metrics to identify whether the failures happen only for specific events or during high traffic.
I would also check throttling, timeout, downstream services, database connections, and recent changes.
Simple ga gurthupettuko:
Intermittent → Logs → Traffic → Throttle → Timeout → Dependency
Eng: I would correlate intermittent failures with traffic, logs, timeouts, throttling, and dependencies.

14.How do you monitor a Lambda function in production?
I would use CloudWatch metrics and logs to monitor invocations, errors, duration, throttles, and other relevant metrics.
I would configure alarms for important conditions such as high errors or throttling.
Simple ga gurthupettuko:
Lambda → CloudWatch Metrics + Logs → Alarms
Eng: I would monitor Lambda using CloudWatch metrics, logs, and alarms.

15.What is your overall approach to troubleshoot a Lambda issue?
First, I identify whether the issue is related to code, permissions, networking, event input, performance, or a downstream dependency.
Then I check CloudWatch metrics, logs, IAM role, configuration, VPC settings, and recent changes.
Simple ga gurthupettuko:
Event → Code → IAM → Network → Dependency → Metrics → Logs → Root Cause
Eng: I identify the failing layer and use metrics, logs, permissions, and configuration to find the root cause.

API Gateway Troubleshooting — Important Interview Questions & Answers

1.API Gateway is returning 403 Forbidden. How will you troubleshoot?
First, I would check the authorization configuration and whether the client has the required permissions.
Then I would check API Gateway resource policies, IAM authorization, Lambda authorizer, API key requirements, and the requested stage/path.
Simple ga gurthupettuko:
403 → Authorization → IAM → Authorizer → Resource Policy → API Key → Stage
Eng: I would check authorization, resource policies, authorizers, API keys, and stage configuration.

2.API Gateway is returning 401 Unauthorized. What will you check?
I would check whether the request contains the required authentication information.
Then I would verify the authorizer configuration, token, API key if applicable, and authorization settings.
Simple ga gurthupettuko:
401 → Authentication → Token → Authorizer → API Key
Eng: I would check the authentication credentials and authorizer configuration.

3.API Gateway is returning 500 Internal Server Error. How will you troubleshoot?
I would check whether the problem is coming from API Gateway integration or the backend service.
Then I would check Lambda or backend logs, integration configuration, permissions, and recent changes.
Simple ga gurthupettuko:
500 → Integration → Backend → Lambda Logs → Permissions → Recent Changes
Eng: I would check the integration and backend logs to identify the internal error.

4.API Gateway is returning 502 Bad Gateway. What does it usually indicate?
A 502 commonly indicates that API Gateway received an invalid response from the backend integration or could not properly communicate with it.
I would check the integration configuration and backend response, especially for Lambda integrations.
Simple ga gurthupettuko:
502 → Integration Problem → Backend Response → Lambda/API
Eng: I would check whether the backend integration is returning a valid response.

5.API Gateway is returning 504 Gateway Timeout. What will you check?
I would check whether the backend is taking too long to respond.
Then I would check Lambda execution duration, backend processing, database calls, and other downstream dependencies.
Simple ga gurthupettuko:
504 → Backend Slow → Lambda/EC2 → DB/API → Timeout
Eng: I would investigate why the backend is taking too long to return a response.

6.API Gateway works but Lambda is not getting invoked. What will you check?
I would check the API Gateway integration configuration and Lambda invocation permission.
Then I would check the API stage deployment and CloudWatch logs.
Simple ga gurthupettuko:
API → Integration → Lambda Permission → Stage Deployment → Logs
Eng: I would check the integration, Lambda permission, deployment, and logs.

7.API Gateway changes are not reflecting. What could be the reason?
In API Gateway, changes may need to be deployed to the appropriate stage before they become active.
I would verify that the latest API configuration has been deployed to the correct stage.
Simple ga gurthupettuko:
Change → Deploy → Correct Stage → Test
Eng: I would verify that the latest API changes are deployed to the correct stage.

8.API Gateway is throttling requests. How will you troubleshoot?
I would check the API Gateway throttling configuration and identify whether the request rate is exceeding the configured limits.
I would also check usage plans, API keys, traffic spikes, and backend capacity.
Simple ga gurthupettuko:
Throttle → Request Rate → Limits → Usage Plan → Traffic
Eng: I would check throttling limits, usage plans, and traffic patterns.

9.API Gateway endpoint is not reachable. What will you check?
I would first verify the API URL, stage, resource path, and HTTP method.
Then I would check whether the API is deployed and whether DNS or custom domain configuration is involved.
Simple ga gurthupettuko:
URL → Stage → Resource → Method → Deployment → Custom Domain
Eng: I would verify the endpoint, stage, resource, method, deployment, and domain configuration.

10.API Gateway custom domain is not working. What will you check?
I would check the custom domain configuration, DNS record, certificate, and API mapping.
I would also verify that the domain points to the correct API Gateway endpoint.
Simple ga gurthupettuko:
Custom Domain → DNS → Certificate → API Mapping → Endpoint
Eng: I would check DNS, the certificate, API mapping, and endpoint configuration.

11.API Gateway authorizer is failing. How will you troubleshoot?
I would check whether the authorizer is receiving the expected token or request information.
Then I would check the authorizer configuration, Lambda authorizer logs if applicable, IAM permissions, and authorization response.
Simple ga gurthupettuko:
Request → Token → Authorizer → Logs → Permission → Response
Eng: I would check the token, authorizer configuration, logs, permissions, and authorization response.

12.API Gateway returns success but the client gets an incorrect response. What will you check?
I would check the backend response and API Gateway integration response configuration.
For Lambda integrations, I would verify that the Lambda returns the expected response structure, status code, headers, and body.
Simple ga gurthupettuko:
Success → Backend Response → Integration → Status Code → Body
Eng: I would check the backend response and API Gateway integration configuration.

13.How would you troubleshoot API Gateway performance issues?
I would check API Gateway latency, integration latency, error rates, throttling, and backend response time.
Then I would check the backend service such as Lambda, EC2, or database for performance issues.
Simple ga gurthupettuko:
API Latency → Integration Latency → Backend → DB
Eng: I would compare API and integration latency to identify whether the issue is in API Gateway or the backend.

14.How would you troubleshoot an API Gateway issue in production?
First, I would identify the exact HTTP status code and affected API, stage, resource, and method.
Then I would check API Gateway logs, metrics, authorizer, integration, backend logs, throttling, and recent deployments.
Simple ga gurthupettuko:
Status Code → API → Stage → Method → Logs → Integration → Backend
Eng: I would identify the failing API path and trace the request through authorization, integration, and backend.

15.What is your overall approach to troubleshoot API Gateway?
I would follow the request flow from the client to API Gateway, then to authorization, integration, and backend.
I would check the status code, stage, method, authorizer, throttling, integration, and backend logs.
Simple ga gurthupettuko:
Client → API Gateway → Authorizer → Integration → Lambda/Backend → Response
Eng: I would follow the complete request flow and identify the layer where the failure occurs.
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